order10 RESULTS: Laplacian of the weight-0 Siegel lift Φ_0 on the unit-arc boundary rays
Prereg: PREREG.md, sha256 f5d2213d…c24e in PREREG.sha256, written before any numerics. Scripts are hashed in scripts/SHA256SUMS.

Theorem (Category A, derived in PREREG §1 before numerics)
$\Delta_0\theta_z = \tfrac{1}{4}\theta_z - \tfrac{1}{4}\partial_\mu^2\theta_z$, therefore
$(\Delta_0 - \tfrac{1}{4})\Phi_0 = (\pi/2)\cdot y^{3/2}\cdot[B(\gamma\hat{c}1) - B(\hat{c}1)]$,
with $B(\hat{c}1) = \sqrt{140}\cdot\vartheta^+(\tau)\cdot\overline{\vartheta^-(\tau)}$. The orientation sign is +1 (measured).
- $\vartheta^+ = \Sigma_{m\in\alpha1+\alpha2+Z}\, m\cdot e(7m^2\tau/2 + 7(b0+b1)m)$: weight-3/2 holomorphic, on the ray e2.
- $\vartheta^- = \Sigma_{u\in\alpha1+Z}\, u\cdot e(5u^2\tau/2 + 5b0\cdot u)$: weight 3/2, the $5n^2/2$ series, on the negative line c1.
- The $\gamma\hat{c}1$ endpoint (ray γe2) has the same form with $\alpha \to \gamma^{-1}\alpha$.

Verdicts (Category B numerical confirmation, mpmath dps 30)
Test	Result	Verdict
D1: per-Fourier-mode identity, $\|N\| \le 8$, $y \in \{\frac{1}{2}, 1, 2\}$, 18 companions	max relative error 7.1e−30	**pass**
D2: exact factorisation into $\vartheta^+\cdot\overline{\vartheta^-}$, 3 τ	1.0e−30	**pass**
D3: characteristics (exact rationals)	the γ-end $5n^2/2$ characteristics $(\gamma^{-1}\alpha)_1$ mod 1, up to sign, lie in the same set of 10 classes as $\alpha1$	**pass (exact)**
D4: γ-invariant companions have zero defect	8.5e−7, inconclusive by the preregistered rule	**inconclusive**
R1′: 0_E at x = 1/4, 9 terms on 12 points, dps-40 least squares	max error 1.4e−7	**pass**
R2′: $0_{C_{arc}}$ A0 at x = 1/4	6.3e−12	**pass** (Lemma-2 type, descriptive, not counted)

D4 note. The preregistered rule gives inconclusive. A post-hoc rerun (d4posthoc.py, descriptive only) used the full Q-shell (box 200 instead of 14) and got ≤ 1.0e−29 for all 15 γ-invariant companions. This shows the shortfall was a box-truncation artefact: γ moves shell points outside a 29×29 box. With the full shell, the γ-invariant companions are pure $\Delta = \frac{1}{4}$ eigenforms.

R1′ settles order9's x = 1/4 inconclusive as a fit-resolution artefact; the order9 verdict stays recorded as inconclusive.

Shadow verdict (preregistered rule)
- The boundary terms are located exactly (Category A, confirmed Category B). They are products $y^{3/2}\cdot\vartheta^+\cdot\overline{\vartheta^-}$ of two weight-3/2 unary thetas at each ray.
- **The even-piece and full-series ξ_1-shadow is NOT located.** Φ_0 is a weight-0 $\Delta = \frac{1}{4}$ object with boundary, not a weight-1 completion of 0_E.
- The antiholomorphic factor ϑ^- is the $5n^2/2$ series again. The distinct weight-3/2 unary series in the ξ-sense therefore stay **three** ($5n^2/2$ shared; plus $R^2/34$ and $35m^2/4$).
- The new ingredient is the weight-3/2 holomorphic partner $\Sigma\, m\cdot e(7m^2\tau/2)$. It appeared before only as the weight-$\frac{1}{2}$ $\theta_{7/2}$.
- $0_E = \boxed{A}(\Phi)$ remains an identification, not a bridge. $M^2 = 35\cdot P$ is exact (Category A, order9 cyclo).

Disclosures
- D1 is a fundamental-theorem-of-calculus identity checked per ν on a truncated box. Its strength is the closed form of the boundary term, not independent evidence of a shadow.
- In D2 the twist split $B(\nu,\beta) = -5b0\cdot u + 7(b0+b1)\cdot m$ is exact. The γ-end twist is evaluated numerically.
- R1′/R2′ use float64 lattice sums with a dps-40 normal-equation fit.
- The D4 FAIL/inconclusive was recorded before the post-hoc rerun. The post-hoc result is not counted as a preregistered pass.
- CORE_FREEZE was respected and the 35-component law vector was not touched.

Open next step
Build a weight-1 non-holomorphic completion of 0_E whose ξ_1 is $\Sigma \pm\overline{\vartheta^-}$ at the two rays, an Eichler integral of the $5n^2/2$ series against ϑ^+. Test its S-law against the 105×105 M with exact coefficients. Alternatively, show it is impossible via the P-even parity lemma, which applies only to holomorphic τ-laws (review finding 1).